Chapter 5
A Critical Perspective on Nanoplastics in Soil and Crop Systems

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Abstract Nanoplastics have emerged as a significant emerging contaminant of the Anthropocene. The ubiquitous presence of nanoplastics attracted considerable attention due to their potential impacts on soil and crop systems. The nanoplastics can be primary and secondary in origin. Their interaction and fate in soil depend on the physicochemical properties, including size, shape, surface charge, hydrophobicity and polymer type. Nanoplastics can both adsorb and leach contaminants, altering soil chemistry and increasing the bioavailability of toxic substances to plants and microorganisms in the rhizosphere. Nanoplastics interferes with soil-plant-microbe interactions, often bringing changes in rhizosphere microbial community composition and function, negatively impacting soil health and crop productivity. Experimental evidence on nanoplastics uptake by roots exacerbates their accumulation in other tissues, including edible plant parts, posing risks to food safety and human health.

Nanoplastics are reported to induce oxidative stress in plants and soil microorganisms, impairing key physiological processes and reducing crop yields. Reports of nanoplastic bioaccumulation within the food chain underscore the need for risk assessments. Despite a growing body of evidence, significant knowledge gaps remain, especially in the detection, quantification, and characterisation of nanoplastics in the environment. Even though pyrolysis GCMS can help identify the polymer, the approach provides no information on the shape, number and size of the nanoplastics, which is a severe constraint in assessing nanoplastic contamination

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and toxicity, which is size dependent. Significant efforts are needed before any realistic ecological impact assessment can be undertaken to understand their long-term environmental and health implications better.

Keywords Pyrolysis-GCMS · Nanoplastics · Oxidative stress · Rhizosphere

5.1 Introduction

With the progress in research on microplastics (<5 mm in size), there are increasing efforts to understand the fate of even smaller plastic particles: the nanoplastics (Gigault et al. 2016). The upper size limit of nanoplastics is still being discussed, with the literature variously suggesting either 100 nm or 1000 nm (Gigault et al. 2018a, b; Li et al. 2020a; Alexander et al. 2016; Ng et al. 2018). A reliable report of the likely presence of nanoplastics in the environment was first put forward by Gigault et al. (2016) suggesting that microplastics can fragment under solar radiation. The presence of nanoplastics within soil and crop systems has emerged as a critical issue (Atugoda et al. 2023) as they potentially present a more insidious form of contamination for multiple reasons, i.e. their enhanced toxicity, their colloidal behaviour and stability under a variety of environmental conditions, ability to move across cell membranes, and most importantly, the difficulties in isolating and identifying them in environmental samples.

Due to small size and unique physicochemical properties (Gigault et al. 2021; Covernton et al. 2019), nanoplastics pose distinct challenges to soil health (Wu et al. 2020; Hurley and Nizzetto 2018), crop productivity, and human health. They can originate from a variety of sources, both primary and secondary. Primary nanoplastics are intentionally manufactured at the nanoscale for use in industrial applications (Stephens et al. 2013; Rodríguez-Hernández et al. 2020), including abrasives, inclusions in cosmetics (Hernandez et al. 2017), fillers in coatings, and drug delivery systems (Marchesan and Prato 2013; Shi et al. 2015; Ali et al. 2020). These particles often enter soil systems through wastewater irrigation, land application of sewage sludge (Frehland et al. 2020), flooding, and atmospheric deposition. Secondary nanoplastics, on the other hand, are formed through the degradation of larger plastic debris in the environment (Gigault et al. 2021), and one of the conspicuous sources of smaller plastics is mulch (Seo et al. 2024; Nizzetto et al. 2016; Bao et al. 2024; Ferreira-Filipe et al. 2024). Photodegradation, mechanical weathering, and biological activity (Cai et al. 2023) can break down macroplastics and microplastics into nanoplastic particles (Wang et al. 2020; Astner et al. 2019; Zhang et al. 2019). Agricultural practices, such as using plastic mulching films, fertilisers, and pesticide containers, contribute significantly to soil nanoplastics accumulation. Based on the 88–99.9% of microplastic retained in sewage sludge (Uddin et al. 2020, 2022), it can also be a significant source of nanoplastics. However, data on nanoplastic inventories are scarce due to challenges such as matrix interference in their analysis; sewage sludge is rich in organic compounds, dissolved solids and microorganisms (Cavazzoli et al. 2023).

Two agricultural practices are primarily responsible for introducing micro- and nanoplastics into agricultural soil: The pervasive use of plastics in agriculture exacerbates this issue. For instance, plastic mulches, widely used to conserve moisture and control weeds, degrade over time, releasing microplastics and eventually nanoplastics into the soil (Atugoda et al. 2023; Astner et al. 2019; Zhang et al. 2019); and the application of biosolids as fertilisers that transfer the NPs present in sewage sludge directly into agricultural soils (Mclaughlin and Pennock 2018).

5.2 Behaviour of Nanoplastics in Soil Systems

The behaviour of nanoplastics in soil is influenced by their physicochemical properties, including size, shape, surface charge, and hydrophobicity. Due to their small size, nanoplastics have a large surface area-to-volume ratio, which enhances their reactivity and interaction with other soil components. Nanoplastics can adsorb metals on the surface and absorb persistent organics in the bulk (Rochman et al. 2013a, b, 2014), potentially altering the chemical properties of the soil. Additionally, their hydrophobic nature can influence soil water retention and transport processes, affecting soil structure and function.

The interaction of nanoplastics with the microorganisms in soil can result in changes within the microbial community composition and function (Aralappanavar et al. 2024). These interactions can have far-reaching effects on soil fertility and health, as the soil microbes play a pivotal role in maintaining soil structure, nutrient cycling, and decomposition of organic matter (Aralappanavar et al. 2024; Nath et al. 2024). The vertical migration of nanoplastics with the irrigation return flow or infiltration with precipitation can transport them to deeper soil layers and even to underground aquifers. The mobility of nanoplastics in soil is influenced by several factors, such as soil texture, moisture content, and organic matter. Understanding the fate and transport of nanoplastics in soil is essential for assessing their long-term environmental impacts.

Extracting nanoplastics, especially from organic-rich soils, is challenging (Wahl et al. 2021), hence the scarcity of nanoplastic inventories in soils (Hurley and Nizzetto 2018). A detailed review by Wahl et al. (2021) on agricultural soil in France that has been contaminated with plastic debris for about 30 years appears to be a proof of concept, but lacks an inventory of the site. They extracted nanoparticles from soil samples using soil-water extraction with ultra-pure water in a 1:4 ratio, stirred at 300 rpm for 72 hours without any pH adjustment (Wahl et al. 2021). The samples were filtered on 0.8 μm cellulose acetate filters (Wahl et al. 2024), and the water extract was fractionated utilising asymmetric flow-field flow fractionation (AFF) where the accumulation wall was a polyethersulfone (PES) membrane. The use of AFF coupled online to a static light scattering (AF4-UV-SLS) in the study suggests the colloidal populations in two size ranges of 20–150 nm and 150–500 nm, whereas the Py-GCMS was used to identify polymers.

As in the French agricultural soil amended with sewage sludge, NPs are likely ubiquitous in other regions' agricultural farms and soils. In another study, nanoplastics were reported in the soil profile 0–55 cm (Wahl et al. 2024). Ng et al. (2018) computed a lifetime addition of 2.3–63 tons per hectare of microplastics to agricultural soils, which Atugoda et al. (2023) misattributed as a nanoplastic and an annual addition rather than a lifetime. Nevertheless, microplastic degradation may result in nanoplastic. Still, its quantification is impossible as the nanoplastic formation rate will depend on the type of polymer, soil type and conditions favourable to microbial degradation, such as organic matter and moisture content. The microplastic inventories were higher in the topsoil, i.e., 78.00 items kg^{-1} compared to 62.50 items kg^{-1} in deep soils (Liu et al. 2018a). However, Liu et al. (2018a) also indicated that the larger plastics were found in the topsoil while the deeper soils were enriched with smaller microplastic size fractions; with this consideration, should we expect nanoplastics inventories to be higher in the soil beneath? This can partly be due to the movement of NPs through the interstitial spaces in the soil profile, with the irrigation return flow.

5.3 Impact on Crop Systems

The presence of nanoplastics in soil can directly and indirectly affect crop systems. One primary concern is plants' potential uptake of nanoplastics (Li et al. 2021a). The first evidence of nano bead internalisation in crops came in 2012 (Bandmann et al. 2012), where it was reported that 20 nm beads were taken up quickly by BY-2 cells, while the 100 nm size beads were excluded from uptake by turgescent and plasmolysed cells (Bandmann et al. 2012). On the other hand, the protoplasts could internalise 1000 nm size beads. Studies have shown that nanoplastics and smaller MPs can be absorbed by plant roots (Dong et al. 2021; Li et al. 2021b; Liu et al. 2022) and translocated to various plant tissues, including leaves (Dong et al. 2021; Sun et al. 2021), stems (Liu et al. 2022; Sun et al. 2021; Lian et al. 2020a), and fruits (Li et al. 2020b, 2021b).

This uptake can occur through various complex mechanisms and pathways, including uptake by root pores from soil colloids and endocytosis. According to a study (Bandmann et al. 2012), bead internalisation was only partially hindered by Ikarugamycin, a clathrin-specific inhibitor, while wortmannin completely blocked it. This suggests that endocytosis happens through both clathrin-dependent and independent pathways, supporting the view that clathrin-independent endocytosis is widespread in plants.

The distribution and movement of nanoplastics (NPs) in soil are significantly influenced by soil colloidal behaviour, which governs their transport, retention, and aggregation (Liu et al. 2021a). Soil colloids, including clay minerals, organic matter, and metal oxides, interact with NPs through electrostatic forces, van der Waals interactions, and cation bridging, affecting mobility (Adeel et al. 2021). Nanoplastics

adsorb onto colloids, forming heteroaggregates that can lead to immobilisation as larger aggregates are filtered out in the soil matrix (Gigault et al. 2018b; Pradel et al. 2023). Furthermore, environmental factors such as soil pH, moisture content, ionic strength and organic matter levels influence colloidal stability; stable colloids can transport attached nanoplastics deeper into the soil profile, potentially reaching groundwater, while conditions promoting flocculation can result in surface deposition (Xu et al. 2023a). Thus, the dynamic interplay between soil colloidal behaviour and nanoplastic properties is critical for understanding contaminant dispersion (Pérez-Reverón et al. 2023), with significant implications for environmental risk assessment and management practices.

The translocation of nanoplastics within plants raises concerns about their accumulation in edible plant parts, potentially entering the food chain and posing risks to human health. In a study, peat moss (*Sphagnum palustre*) exposed to 50 nm polystyrene beads demonstrated preferential adsorption of large aggregates on the leaf surface. Additionally, single nanoparticles were localised within the hyalocysts, which are empty cells containing pores ranging from 5 to 10 μm in diameter and in the cytoplasm (Capozzi et al. 2018). Nanoplastics ranging in size from 50 to 500 nm were also detected in garden cress (*Lolium perenne*). These nanoparticles obstructed the seed capsule's pores, with a pronounced accumulation on the root hairs (Bosker et al. 2019). These particles were also detected on the leaves and epidermal surfaces (Bosker et al. 2019). In another study, 20 nm polystyrene beads were detected within the midrib and veinlets of *Vigna radiata* (mung bean), with prominent aggregation observed in the veins—indicating nanoplastic translocation from the soil into the plant body (Chae and An 2020). Additionally, nanoparticles were observed along the vessel walls in the exposed leaves (Chae and An 2020); however, the possibility of atmospheric deposition and endocytosis cannot be ruled out.

In another experimental study, 50 nm and 500 nm polystyrene beads were applied to soil planted with *Spirodela polyrhiza* (great duckweed). The nanoplastic beads were predominantly aligned along the root shafts and tips, with a uniform distribution observed across the entire root surface (Dovidat et al. 2020), suggesting their likely uptake into plants. In another experiment using *Allium cepa* (garden onion), uptake of 50 nm polystyrene beads was reported (Giorgetti et al. 2020), where the nanoplastics accumulated within the cellular compartments, including vacuoles and the cytoplasm of the root epidermis, resulting in a reduction in root growth.

The genotoxicity of 100 nm polystyrene beads in *Vicia faba* (Faba bean) was reported; similar to great duckweed, the nanoplastics were found localised within the root tips (Jiang et al. 2019) due to either endocytosis or colloidal transfer, elucidating the potential translocation of nanoplastics from the soil into the plants (Jiang et al. 2019). *Triticum aestivum* (wheat) was exposed to 200 nm and 2.0 μm polystyrene beads in another experimental study. These beads were detected in the stems and leaves, providing evidence of the ability of nanoplastics to move through the plants (Lian et al. 2020b). In an additional experiment by the same group, both *Triticum aestivum* and *Lactuca sativa* (lettuce) were treated with 200 nm and 2.0 μm

beads composed of polystyrene and poly(methyl methacrylate) (PMMA); nanoplastics were detected in the roots, shoots, xylem, and leaf veins, further indicating effective translocation of nanoplastics from soil to the crop. The presence of nanoplastics within different parts of the crops underscores their ability to impact both plant physiology and food safety.

In another study, the barley roots readily absorbed PMMA beads of 96.75 ± 0.58 nm diameter (Li et al. 2021c), suggesting the capability of nanoplastic incorporation into plant roots and likely influence in nutrient uptake and effect on plant health. Studies have shown different behaviours of positive and negative charges on 200 nm polystyrene beads in *Arabidopsis thaliana* (Arabidopsis). The negatively charged beads were mainly localised within the stele of the maturation zone, root tips, and the border cells of the root (Sun et al. 2020). Meanwhile, the positively charged beads were on the root hairs and within intercellular spaces. These results underpin the importance of surface charge in controlling the nanoplastic uptake and distribution in plant tissues. In both *Triticum aestivum* (wheat) and *Arabidopsis thaliana* (Arabidopsis), exposure to 40 nm and 1000 nm polystyrene beads resulted in a pronounced accumulation along the root surface, with a particularly notable concentration at the root tip (Taylor et al. 2020). This evidence, along with other reports, suggests that root tips are the main point of uptake of nanoplastics. The studies included here explicitly highlight the importance of particle size in adhesion uptake and endocytosis, providing significant insights towards the mechanisms of nanoplastic mobilisation from the soil into monocot and dicot plant species. Evidence of enhanced antioxidant activity and activated carbon metabolism has been reported due to the localisation of 20 nm polystyrene beads in the roots of *Oryza sativa* (rice) (Zhou et al. 2021).

Based on the published information, it can be hypothesised with reasonable confidence that nanoplastics in the rhizosphere (Chen et al. 2022a) can impact plant growth and development. The nanoplastics in the soil region surrounding plant roots can alter root morphology, affecting plant growth (Roy et al. 2022). The nanoplastic exposure impedes root elongation and affects root hair growth and nutrient uptake (Khan et al. 2024; Hussain et al. 2025). Such physiological changes reduce overall plant health and lower crop yields (Jia et al. 2023). Nanoplastics are also known to affect soil-plant-microbe interactions, potentially disrupting beneficial symbiotic relationships, such as those with mycorrhizal fungi and nitrogen-fixing bacteria.

Moreover, nanoplastics can interact with other soil contaminants like metals and organics, potentially enhancing their bioavailability and toxicity to plants. For example, nanoplastics can adsorb heavy metals (Xu et al. 2023b; Ceylan et al. 2024) and organic pollutants (Liu et al. 2021a), sorbed by nanoplastics, may facilitate their uptake by plant roots and increase their accumulation in plant tissues (Chen et al. 2022b). This co-contamination can have synergistic effects, exacerbating the toxic effects of pollutants on crops and compromising food safety (Liu et al. 2021b).

5.4 Ecotoxicological Concerns

With the extremely large specific surface area, nanoplastics release additives that are added to plastics manufacturing at a faster rate. The lipophilic additives on leaching can penetrate cell membranes and potentially bioaccumulate (Kwon et al. 2017). Additionally, residual monomers from polymerisation in producing plastics, solvents, and additives may leach out of the plastic matrix, posing chronic toxicological risks to plants and organisms (Atugoda et al. 2023; Omidoyin and Jho 2024; Ramanayaka et al. 2024). There is an immediate need to assess the leaching rates of these additives from different polymers under ecologically relevant environmental conditions; the available information is not comprehensive enough to evaluate the actual risk it may pose.

Hydrophobic contaminants like hydrocarbons, polycyclic aromatic hydrocarbons (PAHs), polybrominated diphenyl ethers (PBDEs), polychlorinated biphenyls (PCBs), and pharmaceuticals (Shen et al. 2019) in the soil or water are adsorbed over nanoplastics due to their high partition coefficient (Sun et al. 2023).

There are only seven studies published on the adsorption of organic contaminants onto nanoplastics of different sizes (Table 5.1). Studies have indicated that PAHs and PBDEs have 2–3 orders of magnitude higher partitioning onto polystyrene nanoplastics than PS microplastics (Velzeboer et al. 2014; Liu et al. 2016); this can be logically attributed to the higher specific surface area-to-volume ratio. In another study, the hydrophobicity of contaminants is stated as a key factor in determining the binding ability of non-polar and weakly polar contaminants to nanoplastics (Liu et al. 2018b), controlling their distribution compared to polar compounds. On the other hand, two chemicals, PCB-77 and PAH benzo[k]fluoranthene, have similar hydrophobicity, but the sorption on PS nanoplastics is much higher for benzo[k]fluoranthene and is explained by stronger π-π interactions (Liu et al. 2016). Higher adsorption of ciprofloxacin (0.41 mg g^{-1}) and bisphenol-A (7.45 mg g^{-1}) on aged nanoplastics (Xiong et al. 2020) indicates enhanced affinity of these contaminants to weathered nanoplastics. The sorption of pharmaceuticals onto nanoplastics can enhance these contaminants' spatial disposition and bioavailability. This co-transport facilitates the widespread distribution of pharmaceuticals beyond their original discharge points, increasing environmental persistence (Alimi et al. 2018). In another study, divalent ions are shown to reduce the sorption of tetracycline on polystyrene nanoplastics by blocking binding sites on the nanoplastic surface, thus preventing interaction with antibiotic molecules (Wan et al. 2019). The sample matrix, pH, salinity, and organic matter influence nanoplastic interaction with contaminants.

A few studies have reported the adsorption of metals onto nanoplastics. Davranche et al. (2020) found high Pb(II) levels on nanoplastics processed in the laboratory from plastic particles extracted from the North Atlantic Gyre. Eleven studies have examined nanoplastic and metal/metalloid effects on the end receptors, including terrestrial and non-terrestrial biota. The studies summarised in Table 5.2 demonstrate the potential for nanoplastics to interact synergistically with various

Table 5.1 Organic Contaminants Adsorbed on nanoplastics of different sizes

Polymer	Size in nm	Contaminants	Adsorption mg/g	References
Polystyrene	70	Polychlorinated biphenyls 180	6.61×10^{133}[a, b]	Velzeboer et al. (2014)
		Polychlorinated biphenyls 126	2.69×10^{76}[b]	
		Polychlorinated biphenyls 77	1.05×10^{31}[b]	
		Chrysene	6.8[a, b]	Liu et al. (2016)
		Benzo[a]pyrene	33.8[b]	
		Benzo[b]fluoranthene	42.6[b]	
		Benzo[k]fluoranthene	33.1[b]	
		Benzo[a]anthracene	7.9[b]	
		Benzo[g,h,i]perylene	20.0[b]	
	50	Phenanthrene	2[b]	Ma et al. (2016)
	300–400	Tetracycline	44.8[b]	Wan et al. (2019)
Polystyrene (Pristine)	40	Ciprofloxacin	5.6[b], 3.34[c]	Xiong et al. (2020)
		Bisphenol-A	1.26[c]	
Polystyrene (Aged)		Ciprofloxacin	5.4[b], 3.75[c]	
		Bisphenol-A	8.71[c]	
Polystyrene, humic acid, pH6	300–400	Tetracycline	51.0[b]	Wan et al. (2019)
Polystyrene, $MgCl_2$, pH6			37.5[b]	
Polystyrene-COOH	200	Ciprofloxacin	5.2[b]	Yilimulati et al. (2021)
	500		4.3[b], 3.82[c]	

[a]$(mg\ kg^{-1}/\mu g\ L^{-1})^n$
[b]Isotherm modelling
[c]Kinetic modelling

metals, leading to increased toxicity in several organisms. Many of these organisms are not in the soil, but these findings underscore the importance of considering the combined effects of nanoplastics and metals. The paucity of soil data will limit any realistic environmental risk assessments, as their co-exposure can significantly amplify harmful outcomes among the soil microbial community and probably can also affect the crops.

The ecotoxicological impacts of nanoplastics on soil and crop systems are not yet fully understood, but emerging evidence suggests that they can potentially pose significant risks to soil microbiota and crops. Nanoplastics can induce oxidative stress in plants by producing reactive oxygen species (ROS) that cause cellular damage. This stress can impair photosynthesis, reduce growth, decrease crop productivity, affecting germination, flowering, and fruiting in plant's. The potential for nanoplastics to accumulate in the food chain is also a realistic concern. If nanoplastics are taken up by crops, they could be transferred to herbivores and, subsequently, to higher trophic levels, including humans. The long-term health implications of

Table 5.2 Experimental data on polystyrene nanoplastic and metal toxicity on biota

Size in nm	Metal	Significant findings from treatments of coexposure	References
10	Cadmium	The concurrent exposure to nanoparticles (NPs) and cadmium (Cd) markedly accelerated mortality in *Ctenopharyngodon idellus* (grass carp), with varying degrees of tissue damage observed in the gills, liver, kidneys, and muscles—most severe at concentrations of 20 mg L^{-1} Cd combined with 300 μg L^{-1} NP. Additionally, treatment with NPs alone at 30 μg L^{-1} led to increased superoxide dismutase (SOD) activity in both the kidney and liver. In contrast, peroxidase (POD) activity was elevated exclusively in the kidney.	Chen et al. (2022c)
100, spherical		At 0.05 mg Cd L^{-1} there was a significant reduction in *Euglena gracilis* growth; coexposure enhanced microalgal toxicity with synergistic effects. Metabolic disorders and oxidative stress are the main toxic mechanisms.	Cao et al. (2022)
500, spherical	Copper	Significant increase in toxicity compared to nanoplastic or Cu alone; oxidative stress, cytoplasmic vacuolation and destruction of chloroplasts in *Chlorella* sp. TJ6-5 and *Pseudokirchneriella subcapitata* NIES-35	Wan et al. (2021)
87.17–106, spherical		No adsorption of copper ions was detected on PS-COOH nanoparticles. However, changes in their surface charge and hydrodynamic diameter indicate the formation of an eco-corona and a subsequent loss of the cell's native morphology at an exposure level of 10 mg PS L^{-1} in the presence of copper. Furthermore, the nanoparticles' ability to interact with exudates from *Raphidocelis subcapitata* and adhere to its cell wall suggests that their transfer through the food chain is plausible.	Bellingeri et al. (2019)
50 & 55 spherical		Exposure suppressed the growth of *Platymonas helgolandica* var. *tsingtaoensis*, intensifying inhibition as exposure duration and concentration increased. Oxidative damage was evident at PS-NP concentrations exceeding 50 mg L^{-1} and Cu levels of at least 1.5 mg L^{-1}, with the combined exposures exhibiting a synergistic effect.	Gao et al. (2022)
100, spherical	Zinc	At environmentally relevant concentrations, nanoplastics and zinc do not exhibit a combined interactive effect on freshwater microeukaryotic communities; however, zinc alone induces a discernible alteration in community structure. In contrast, nanoplastics appear to have no significant impact. Zinc exposure leads to a marked initial reduction in phototrophic microorganisms, though photosynthetic activity eventually recovers, accompanied by a shift in the dominant taxa from bacillariophytes to chlorophytes.	Sieber et al. (2022)
	Arsenic	Consequent to non-environmental concentrations (1 mg/L), there was an interactive effect of nanoplastics and arsenic; fresh weight and growth rates in *Myriophyllum verticillatum* were significantly reduced. However, the negative effects at lower concentrations were mainly due to arsenic exposure; *Myriophyllum verticillatum* responded to stress by regulating amino acids and organic acids.	Li et al. (2023)
		The synergistic effect of nanoplastics and arsenic was higher than their single toxicity on *Vallisneria denseserrulata* and *Potamogeton crispus*. The nanoplastics were observed in the leaf cells of these submerged macrophytes.	Tang et al. (2022)
	Lead	In zebrafish intestinal cells, co-exposure to nanoparticles at 20 and 200 μg L^{-1} alongside 50 μg L^{-1} lead demonstrated that the 200 μg L^{-1} nanoparticle treatment had the most pronounced impact on macrophages, whereas lead exposure predominantly affected enterocytes.	Yu et al. (2022)
50, 200, 500 Spherical	Gold	Smaller nanoparticles were observed to penetrate both developing zebrafish embryos and their chorion. Exposure to nanoplastics resulted in modest alterations in survival, hatching rates, developmental abnormalities, and cell death. The synergistic effects were notably more pronounced with increasing dose and decreasing particle size. Signs of oxidative stress and mitochondrial damage were also reported.	Lee et al. (2019)
1000 spherical	Chromium	NP ~1820/ml and Cr(VI) at 2 and 5 ppm for 72 h showed low mortality of *Daphnia magna* at NP13.3%, 2 ppm Cr(IV) individually, but the synergistic effect of two led to 74.5 mortality, and when Cr concentration was 5 ppm absolute mortality was observed; Microbial diversity was changed significantly.	Varg et al. (2022)

chronic exposure to nanoplastics through food consumption are still largely unknown, but the potential for bioaccumulation and biomagnification warrants serious consideration (Petersen et al. 2023; Miller et al. 2020; Gupta et al. 2022).

5.5 Knowledge Gaps, Research Needs and Concerns

Despite the growing body of evidence on the presence and effects of nanoplastics in soil and crop systems, significant knowledge gaps remain. The lack of standardised methods to detect and quantify nanoplastics in environmental matrices hinders our ability to accurately assess their prevalence and impacts. The method used for NP extraction from soil samples is usually soil-water extraction. It is reported that even the least contaminated water source (Milli-Q water) has 4.7–64.4 MP particles ml^{-1} (Jones et al. 2024). This significant contamination can be introduced during sample processing, and most published reports have not accounted for such contaminations. The samples are then segregated by filtration, often on the filter with nanosize pores; usually, 0.8 μm pore filters have been used. The efficacy of removing NPs from these filter membranes is not apparent, especially as the filter material is not always inorganic or metallic. The technique primarily used for identifying and characterising NPs is pyrolysis gas chromatography with mass spectrometry (pyro-GCMS). The analyses by pyro-GCMS provide pyrolyzate markers for different polymers (Table 5.3), which have been defined in detail by Matsui et al. (2020), where the m/z signal and retention index (RI) for each polymer product were taken into consideration and summated mass spectra (SMS) were constructed. The rationale for using the retention index for peak identification emanates from the fact that, unlike retention time, which depends on the length of the column, thickness of film, diameter, and size of the transfer line (Melo et al. 2017), the RI is independent of the GC parameters. There are discussions regarding the non-inclusion of styrene since pyrolysis of PS and ABS results in its detection; hence, it should not be used as a characteristic product for identifying polymer species.

The complex interactions between nanoplastics and soil constituents, including organic matter, minerals, and microbes, are not yet fully understood. Long-term studies also need to evaluate the chronic effects of nanoplastics on soil health, crop productivity, and food safety. Research on the potential for nanoplastics to interact with other environmental stressors like thermal and microbial degradation, and changes in soil pH, is vital. Developing sustainable agricultural practices that minimise plastic use and reduce nanoplastics input into the terrestrial environment is crucial for mitigating their impacts.

A significant limitation of the pyro-GCMS is its inability to provide information on the size, count, and shape of the nanoplastics in a sample. The nanoplastic size is critical in assessing their toxicity; the higher surface-to-volume ratio of smaller particles has increased potential toxicity. The smaller nanoplastics have enhanced capability to penetrate biological membranes (Khan et al. 2024), but they also translocate and accumulate in tissues away from their point of contact or entry, leading

Table 5.3 The pyrolysis product of different polymers, with their m/z signals and retention index (After Matsui et al. (2020))

Pyrolyzate	Retention index	m/z	Polymer
1-Undecene	1091	41, 55, 70, 97, 111, 154	Polyethylene
1-Tetradecene	1393	41, 55, 97, 125, 196	
1-Eicosene[a]	1994	43, 55, 83[a], 97, 280	
2,4-Dimethyl-1-heptene[a]	843	39, 41, 43, 55, 70[a], 83, 126	Polypropylene
2,4,6-Trimethyl-1-nonene	1082	41, 43, 55, 69, 83, 111, 125, 168	
2,4,6,8-Tetramethyl-1-undecene (isotactic)	1311	41, 43, 55, 69, 111, 125, 154, 167	
2,4,6,8-Tetramethyl-1-undecene (syndiotactic)	1330	41, 43, 55, 69, 111, 125, 154, 167	
3-Butene-1,3-diyldibenzene	1761	65, 91, 104, 115, 130, 193, 208	Polystyrene
5-Hexene-1,3,5-triyltribenzene[a]	2514	65, 91[a], 117, 194, 207, 312	
Styrene monomer	8.5 min[b]	104	
Toluene	7 min[b]		
Benzene[a]	660	39, 50, 51, 52, 77, 78[a]	Polyvinyl chloride
Naphthalene	11.3 min[b]	128	
naphthalene-1-methyl	12.3 min[b]	142	
Methyl methacrylate[a]	711	39, 41, 59, 69, 99, 100[a]	Poly(methyl methacrylate)
ε-Caprolactam[a]	1276	30, 42, 55, 56, 84, 85, 113[a]	Nylon 6
N-(5-Cyanopentyl)hex-5-enamide	1943	30, 41, 55, 69, 100, 114, 154, 168, 208	
6-Acetamido-N-(5-cyanopentyl) hexanamide	2660	30, 43, 55, 114, 167, 224, 267	
Cyclopentenone[a]	799	39, 41, 42, 55, 56, 84[a]	Nylon 66
Adiponitrile	1200	39, 40, 41, 51, 52, 54, 55, 68	
1,8-Diazacyclotetradecane-2,7-dione	2328	30, 41, 55, 98, 112, 140, 198, 226	
4-Vinylcyclohexene	841	39, 54, 79, 80, 93	Acrylonitrile butadiene styrene
2-Methylene-4-phenylbutanenitrile	1363	39, 65, 91[a], 92, 157	
4-Phenylpent-4-enenitrile	1446	77, 91, 103, 115, 117, 130, 142, 157	
2-Methylene-4-phenylheptanedinitrile	1873	91, 104, 117, 144, 182, 210	
2-Phenethyl-4-phenylpent-4--enenitrile[a]	2234	91[a], 105, 115, 118, 156, 170, 261	

(continued)

Table 5.3 (continued)

Pyrolyzate	Retention index	m/z	Polymer
Dimethyl terephthalate[a]	1524	50, 76, 103, 120, 135, 163[a], 194	Polyethylene terephthalate
Bisphenol A dimethyl ether[a]	2110	133, 211, 241[a], 242, 256	Polycarbonate
4-(4-Aminobenzyl)-N-methylaniline[a]	2282	106, 120, 182, 196, 211, 212[a]	Polyurethane

[a]Compounds used to calculate the signal-to-noise ratios by extracting the m/z profiles of the signal
[b]Retention time

to amplified toxicological effects. The other factor that enhances the toxicity of smaller nanoplastics is their enhanced reactive surface, which binds other hydrophobic contaminants and acts as a vector for them.

The limited information on the penetration capabilities of nanoplastics hinders the development of simulation models to predict their distribution and accumulation in crops and microorganisms realistically. Such models are key to understanding chronic and acute health risks associated with nanoplastic exposure. In the absence of nanoplastic inventories and size information, it is challenging to develop accurate models that can predict the potential health impacts of these particles over time. Elevated nanoplastic concentrations in soil can lead to more significant exposure for microorganisms and, in turn, can compromise crop vitality.

The pyro-GCMS is a potent tool for identifying polymers, but its inability to provide quantitative size and morphological information on nanoplastics complicates the exposure assessment of soil microbes and invertebrates, particularly in terms of nanoplastic effect at the molecular and cellular level. The toxicological outcomes and capability for long-term monitoring and assessment in the absence of such size and count information have high uncertainties. Also, the data generated is often incomparable as it is usually reported in terms of the concentration of polymer per unit mass rather than the number of particles per unit mass or area reported for microplastics.

The experimental toxicological data on nanoplastics is often shown to be size-dependent. The surface area to volume ratio of nanoplastics is a key determinant of their reactivity and potential to be transport vectors for contaminants of concern. Smaller nanoplastics induce higher oxidative stress and inflammation in biological systems (Zhou et al. 2023), increasing toxicity (Geng et al. 2023; Lee et al. 2024; Jinadasa et al. 2023; Uddin et al. 2023).

5.6 Conclusion

Nanoplastics are a significant concern of the Anthropocene, posing an emerging challenge to the soil and crop systems. The nanoplastics have potentially far-reaching consequences for long-term soil health, crop vitality and productivity, and food safety issues. Nanoplastic research is still in its initial stages due to

technological limitations in identifying them in environmental samples. The available evidence, primarily based on experimental data, underpins the significant soil-plant interactions, negative crop growth, and food safety concerns emanating from their presence in various plant tissues. A multidisciplinary team of environmental scientists, agronomists, toxicologists, and policymakers must address this multifaceted challenge. To develop effective management and mitigation strategies for nanoplastic risks in soil and crop systems, data on the sources, behaviour in soil, and response to microbial interactions are needed.

The technical limitations of the pyro-GCMS in providing size and count (numbers) information in environmental samples hamper our ability to assess toxicity accurately. Moreover, in the absence of size information, the surface-to-volume ratio of nanoplastics, its reactive surface and toxicity are questionable. Since risk assessments typically rely on detailed information about the particle's physicochemical characteristics, the lack of such details compromises the efficacy risk assessment. The other concerns emanate from the purity of the media, laboratory blanks, and lack of standard reference material. Most researchers have not reported background concentrations; hence, the data cannot be corrected.

To complement the information on polymer types generated by pyro-GCMS and to overcome the limitations, it is prudent to explore alternative and complementary techniques like dynamic light scattering (DLS) and nanoparticle tracking analysis (NTA) that can provide information on the nanoplastic size distribution and concentration in a sample. It will be interesting to assess how these techniques can be used to obtain a more comprehensive understanding of the characterisation of nanoplastics in environmental samples and to evaluate their realistic toxicological implications.

Furthermore, the inability to assess the size and number of nanoplastics limits the ability to establish exposure-response relationships, which are critical for determining safe levels of exposure and developing regulatory guidelines. Without this information, it is challenging to develop comprehensive risk assessments that fully account for the potential health impacts of nanoplastics. Also, the data produced is incomparable to most of the data on microplastics in published literature. It is prudent to mention that a large volume of literature on nanoplastics has inappropriately combined the information on micro-nanoplastics to the extent that some information is inaccurate and misleading; we have highlighted some of these in this perspective.

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